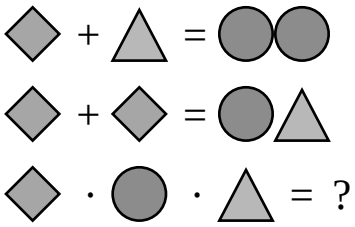
Math Kangaroo Practice

Grade 3-4 · 12 questions · standard · seed 87832

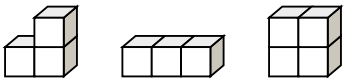
Part A: Each question is worth 3 points

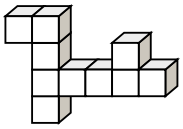
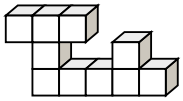
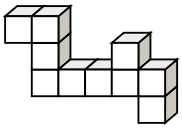
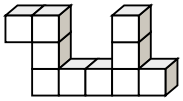
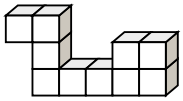
1. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Mia lays out the cards so that the calculations are correct. What number does Mia get for the last calculation?



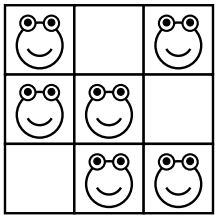
A. 28 ***B.*** 29 ***C.*** 32 ***D.*** 33 ***E.*** 56

2. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures **can** he make?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

3. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Only **one** frog jumps. How many different jumps are possible? (Count every frog and every empty field it can reach.)



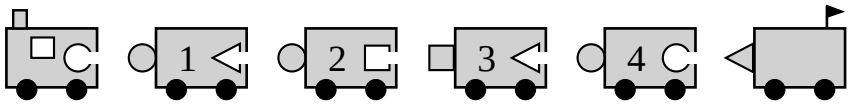
A. 3 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 9

4. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Ida takes all the **heart-shaped** cookies that are still on the tray.
Emma takes all the **white** cookies that are still on the tray.
Lena takes all the **large** cookies that are still on the tray.
First Emma goes, then Ida, then Lena. How many cookies does **Lena** take?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 7 ***E.*** 16

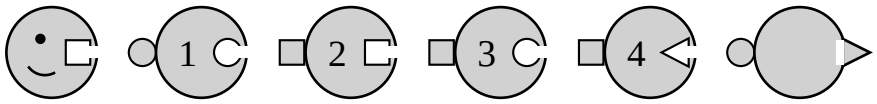
Part B: Each question is worth 4 points

5. Ben has these pieces: an engine, a last wagon and 4 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3 or 4 wagons, then the last wagon. Each piece can be used only once. How many different trains with **exactly 2** wagons can Ben build?



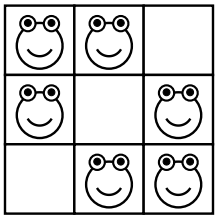
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

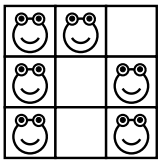
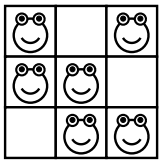
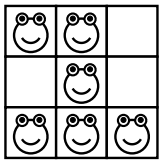
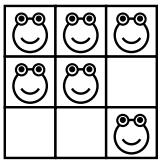
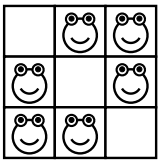
6. Lucas has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 parts, then the tail. Each piece can be used only once. How many different caterpillars with **exactly 2** parts can Lucas build?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

7. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Two frogs jump at the same time, each to an empty neighbouring field. After that there are again 2 frogs in each row and in each column. Which picture can show the pond **after** the jump?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

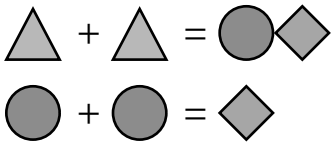
8. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Lena takes all the **heart-shaped** cookies that are still on the tray.
Tina takes all the **white** cookies that are still on the tray.
Zoe takes all the **large** cookies that are still on the tray.
We do not know in which order they go. How many cookies are still on the tray after all three girls have taken theirs?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

9. 60 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: black, striped, black, striped, ... The cards repeat like this: star, diamond, circle, star, diamond, circle, ... What does child number **41** wear?

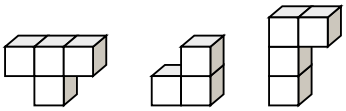
A. black T-shirt, diamond card ***B.*** black T-shirt, circle card ***C.*** striped T-shirt, diamond card
D. black T-shirt, star card ***E.*** striped T-shirt, circle card

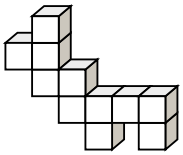
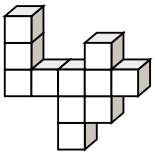
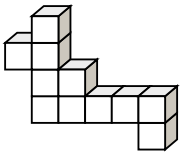
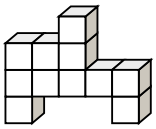
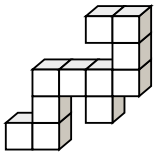
10. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Anna lays out the cards so that the calculations are correct. Which two-digit number do the cards triangle and circle make?



A. 16 ***B.*** 61 ***C.*** 62 ***D.*** 71 ***E.*** 74

11. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures can he **not** make?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part C: Each question is worth 5 points

12. 40 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: black, dotted, striped, white, black, dotted, ... The cards repeat like this: triangle, diamond, triangle, diamond, ... How many children have a **white** T-shirt **and** a **diamond** card?

A. 5 ***B.*** 10 ***C.*** 11 ***D.*** 12 ***E.*** 20

Answer Key

1. **A**	**2. E**	**3. D**	**4. A**	**5. B**	**6. C**
7. **B**	**8. D**	**9. A**	**10. B**	**11. A**	**12. B**

Solutions

1. A (28) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: circle = 1, diamond = 7, triangle = 4. So 7 · 1 · 4 = 28. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · product_all]

2. E — Every figure has 10 cubes, so count is not enough. Start with a cube that sticks out: only one part can fill it. Only one figure can be filled completely. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · can_make]

3. D (8) — Go frog by frog and count its empty neighbours: 1 + 2 + 1 + 2 + 1 + 1 = 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog]

4. A (0) — Cross out what the girls before Lena take first. Of her cookies only 0 are still on the tray (she would get 7 if she went first). [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · given_order]

5. B (2) — Start at the engine and try every piece that fits, one branch at a time (H = engine, T = last wagon): H–2–3–T, H–4–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · exact_length]

6. C (2) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–2–3–T, H–3–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · exact_length]

7. B — First count the frogs in every row and column of each picture. Then look at which frogs moved: in the right picture each of the two frogs moved just one field (not diagonally), and the second jump undoes the first. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · which_picture]

8. D (3) — A cookie stays only if it is not a heart, not white and not large: the small dark round ones. There are 3 of them - the order does not matter. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · left_over]

9. A (black T-shirt, diamond card) — The T-shirts repeat every 2: 41 = 20 × 2 + 1, so the T-shirt is the 1st one: black. The cards repeat every 3: 41 = 13 × 3 + 2, so the card is a diamond. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

10. B (61) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: diamond = 2, circle = 1, triangle = 6. So the triangle is the tens digit and the circle the ones digit: 61. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · plus_minus]

11. A — Try to fill each figure with the parts, starting at a corner or a cube that sticks out. Four figures can be filled; in figure A there is always a gap that no part fits. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · cannot_make]

12. B (10) — Write the first children down: both patterns start again together already after 4 children (not 4 × 2 = 8). In each block of 4 child number 4 matches. 40 ÷ 4 = 10 blocks: 10. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · never_together]